

Agile Methodology

- what is agile?

Agile is set of principles and practices for software development that focus on developing flexibly, incrementally, and with continuous customer feedback.

- Agile development is adaptive not predictive

- Agile four core values,

- ✓ Individual and interaction
- ✓ Working software
- ✓ Customer Collaboration
- ✓ Responding to change

- **Principles of Agile Development**

- satisfaction through early and continuous delivery
- Welcome changing requirements, even late in development
- Deliver working software frequently
- Business-people and developers work together daily
- Build projects around motivated individuals
- Face-to-face communication is most effective
- Working software is the primary measure of progress
- Maintain a constant development pace (sustainable development)
- Continuous attention to technical excellence
- Simplicity – maximize the amount of work not done
- Self-organizing teams produce better designs
- Regular reflection and improvement (retrospectives)

- **What is Scrum**

Scrum is a Agile Frame work

- **Scrum Basic**

Scrum Team

1. Product Owner-manage and priorities the product backlog, clarify requirements, represent the customer, accept or reject completed work.
2. Scrum Master-remove obstacles, facilitate events, coach the team, protect it from interruptions
3. Developers-programmers, UI/UX designers, testers, database developers, DevOps and security engineers

Scrum Event

- 1 .Sprint-A fixed time-box (usually 1–4 weeks) in which selected work is Completed
2. Sprint Planning-Decide what can be delivered, how the work will be Done, and the sprint goal
4. Daily Scrum-A short daily meeting for developers to inspect progress and adjust the plan
4. Sprint Review-Demonstrate the completed increment to stakeholders
5. Sprint Retrospective-Discuss and agree process improvements for the next sprint.

Scrum Artifacts

1. Product Backlog-A prioritized list of everything needed in the product: features, bug fixes, and technical and security improvements, research tasks.
2. Sprint Backlog-The work selected for the current sprint. Sprint 1: student login, password reset, student dashboard, basic profile view.
3. Increment-The completed, usable part of the product produced in the sprint at the end of Sprint 1 students can log in and see their dashboard.

